3. App Flow — User and Admin Experience

Project: Advanced Personal Portfolio Website + Admin Panel **Version:** 1.0

> **Roman Urdu summary:** Is document mein har screen, har click aur har error case ka step-by-step flow hai — visitor ke liye bhi aur admin ke liye bhi. AI builder ko yeh batata hai ke “jab user yeh kare to kya hona chahiye”.

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1. Sitemap

```
Public
/                 Home (one-page sections)
  #home  #about  #skills  #projects  #services  #experience  #testimonials
#faq  #contact
/projects         Full gallery (filters, search, sort, pagination)
/projects/:slug   Case study
/resume           CV page + PDF download
/blog             (flag) Blog listing
/blog/:slug       (flag) Article
/privacy          Privacy policy
*                 Custom 404

Admin (noindex, auth required)
/admin/login
/admin                      Dashboard
/admin/projects  /new  /:id
/admin/categories
/admin/skills
/admin/experience
/admin/services
/admin/testimonials
/admin/blog  /new  /:id
/admin/intro
/admin/media
/admin/settings
/admin/navigation
/admin/seo
/admin/inbox  /:id
/admin/analytics
```

```
/admin/sample-data
/admin/users
/admin/audit-log
/admin/backup
```

2. Global Behaviours (apply to every flow)

- **Loading:** skeletons only where real content is loading; never fake loaders.
- **Empty:** every list has a helpful empty state with a next action.
- **Error:** inline, specific, with a Retry button; the page never goes blank.
- **Focus:** on route change focus moves to the page `<h1>`; a live region announces the new page title.
- **Scroll:** route change scrolls to top (or to the `#anchor`); back/forward restore previous scroll position.
- **Theme:** respected on every screen; no flash.
- **Data mode badge:** in demo mode a small “Demo data” badge appears; sample content has a subtle “Sample” marker only inside admin (public site stays clean unless `sample_banner_enabled`).

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3. Visitor Flows

Flow V1 — First visit: AI intro → cinematic reveal

Step	Visitor sees / does	System does
1	Opens `/`	State `checking-preference`: reads `pf:introSkip`, `prefers-reduced-motion`, admin flag `intro_enabled`
2	Sees dark intro screen with poster instantly	State `intro-loading`; video preloads metadata; 6 s timeout starts
3	Video starts muted; captions visible if enabled	State `intro-playing`; shows Skip Intro, Pause, Unmute, Captions controls
4a	Watches to the end	`ended` event → `transitioning` (once)
4b	Clicks **Skip Intro**	Same transition but shorter (≈ 0.6 s)

Step	Visitor sees / does	System does
5	Sees last frame scale/fade/mask into hero with light sweep	Overlay animates; site underneath is already rendered and `inert`
6	Navbar slides in, then heading, text, CTAs	Staggered reveal; focus moves to hero `<h1>`; overlay unmounted
7	Scrolls and uses the site normally	State `portfolio-ready`; intro never replays on internal navigation

Alternative paths

Situation	Result
Autoplay blocked	`intro-fallback`: poster + large Play button + “Enter site”. Tapping Play starts the video with sound only if the user taps Unmute.
Video error / unsupported / 6 s timeout	Fallback screen, then user (or auto after 3 s) enters the site. Never a dead screen.
Reduced motion enabled	No cinematic effect; quick fade directly to the site; video offered only on Replay.
Visitor previously chose “skip on future visits”	Site opens directly; a small “Replay intro” link remains in footer/menu.
localStorage blocked	Defaults apply (intro plays); no crash.
Visitor presses Skip at the same moment video ends	Guard ensures a single transition.
Admin disabled the intro	Site opens directly.
Mobile portrait	Mobile video source/crop is used; controls sit in safe areas, never over the face.

Flow V2 — Replay intro

1. Visitor clicks **Replay Intro** (footer, mobile menu, command palette).
2. Page scrolls to top, overlay re-mounts with fresh state (no duplicate listeners), video restarts from 0.

3. On end/skip the same transition runs; preference is unchanged.

Flow V3 — Navigate the one-page Home

1. Click a nav link (e.g. **About**) → smooth scroll to `#about` with sticky-header offset; URL hash updates; the About link becomes active (scroll-spy).
2. From another route (e.g. a project page) clicking **Services** navigates to `/#services` and scrolls there after load.
3. Logo click → scroll to top of Home.
4. Scroll progress line fills; the navbar compacts after ≈ 80 px; optional hide on scroll-down/show on scroll-up.
5. **Mobile:** hamburger opens a full-height drawer with focus trap; Escape, outside tap, close button or link click closes it and returns focus to the hamburger.

Flow V4 — Explore projects

1. From Home hero **Explore My Work** → scroll to Featured Projects (or open `/projects`).
2. On `/projects`: choose a **category chip** → list filters instantly; URL updates (`?category=ai`) so results are shareable.
3. Type in **search** (debounced 250 ms) → matches title, description, tech tags; "Showing 3 of 12" text updates; clear (×) button resets.
4. Change **sort** (Featured / Newest / Category).
5. If no matches → empty state: "No projects match 'xyz'. Clear filters" with a working button.
6. More than page size → Load more (focus moves to first newly loaded card).
7. Click a card (whole card is one link to the case study; Live Demo/Source are separate buttons that don't trigger the card link).

Flow V5 — Read a case study

1. `/projects/:slug` loads; skeleton for the hero media only while loading.
2. Reads sections in order: summary, cover/demo, problem, approach, role, process, gallery, tools, challenges, results, links.
3. Click a gallery image → lightbox (focus trap, arrow keys, Escape, swipe on touch, alt text announced) → close returns focus to the thumbnail.
4. Click **Share** → native share sheet if supported, else "Link copied" toast.
5. Use **Previous/Next project** or **Related projects**.

6. Click **Discuss a similar project** → `/#contact` with the project title pre-filled in the message hint.
7. Invalid or unpublished slug → custom 404 with links to `/projects` and Home.

Flow V6 — Services → inquiry

1. Browse service cards; expand details (deliverables, process).
2. Click **Discuss this service** → scroll to Contact with that service **pre-selected** in the dropdown.
3. FAQ accordion: keyboard operable (Enter/Space), one or multiple open.

Flow V7 — Contact form submission

Step	Behaviour
1	Visitor focuses a field → `form_start` analytics event (once).
2	Fills name, email, service, optional budget/timeline, message, ticks consent (if shown).
3	On blur/submit, client validation runs; errors appear inline under each field, error summary receives focus on submit.
4	Valid → button shows spinner + "Sending…" and is disabled; Turnstile token attached.
5a	**Success** (`ok:true` from server) → form replaced by confirmation ("Thanks, I'll reply within <X> days" — text from admin). Reset after 10 s or via "Send another".
5b	**Validation error from server** → field errors mapped inline.
5c	**Rate-limited** → "Too many attempts, try again in N minutes".
5d	**Network/server error** → "Couldn't send. Your message is still here — try again or email me directly" with Retry button and mailto link.

Step	Behaviour
5e	**Honeypot filled** (bot) → silently shown success-like state but nothing stored (no hints to bots).
5f	**Demo mode** → honest message "Demo mode: message was not sent".

Other contact actions: **Copy email** (toast "Email copied"), WhatsApp link opens `wa.me` in new tab, social links open in new tab with safe `rel`.

Flow V8 — Resume / CV

1. Open `/resume` → sees timeline, education, certifications.
2. **Download CV** appears only when a PDF is uploaded; click downloads the file; analytics `cta_click:cv`.
3. **Print** uses print stylesheet (light, no animations).

Flow V9 — Theme toggle

Click sun/moon → cycles Light → Dark → System; saved in `pf:theme`; icon + `aria-label` update; no layout jump.

Flow V10 — Command palette (flag)

1. Press **Ctrl/Cmd+K** (or click the search icon).
2. Dialog opens with focus in the input; groups: Navigate, Projects, Actions (Toggle theme, Replay intro, Download CV, Copy email).
3. Type to filter; Up/Down to move; Enter executes; Escape closes and returns focus.

Flow V11 — Blog (flag)

Listing → filter by category/tag, optional search → article with reading time and related posts → share. Missing slug → 404.

Flow V12 — Language switch (flag, only when complete)

Switcher lists only languages marked "complete" in admin; choosing Urdu flips `dir="rtl"` and font; preference saved in `pf:lang`.

Flow V13 — Error and edge cases

Case	Expected behaviour
Offline / API fails on Home	Sections that failed show inline Retry; others render.
Image fails	Neutral placeholder with alt text retained.
JS animation fails	Content remains visible (no `opacity:0` lock).
Section has no data (e.g. no testimonials)	Section is hidden entirely; for testimonials, “My Process” appears instead.
No CV uploaded	Download button hidden.
External link missing	Button not rendered.
Unknown route	Custom 404 with search/links.

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4. Admin Flows

Flow A1 — Login and session

1. Go to `/admin` → not signed in → redirect to `/admin/login` (remember intended page).
2. Enter email + password → **Sign in**. Wrong credentials → generic error “Email or password is incorrect”; repeated failures show a cool-down message.
3. If TOTP is enabled → 6-digit code screen.
4. Success → Dashboard. Idle 60 min → warning dialog “Stay signed in?” → auto sign-out.
5. **Forgot password** → email link → set new password.
6. A signed-in user who is **not** in `admin_users` sees “No access” and is signed out.

Flow A2 — First-run setup (right after the first Owner is created)

1. Dashboard shows a **Launch Readiness** panel and a banner: “You are viewing sample content.”
2. Guided checklist (each item opens its screen):
   - Profile & Settings (name, title, bio, portrait, socials, contact email)
   - Intro video (upload, poster, captions)

- CV PDF
- Add/replace projects (≥ 3 real)
- Skills & tools, Experience
- Services
- SEO defaults and OG image
- Email notification test
- Delete or replace remaining sample data

3. Progress bar updates as items complete.

Flow A3 — Replace sample data with real data

1. Open **Sample Data** screen → grouped list (Projects 6, Skills 18, Services 4 …) each row with “Sample” badge.
2. Per row options: **Edit** (opens editor; saving any change with real text clears `is_sample` after the owner ticks “This is real content”), **Delete**, **Duplicate as new**.
3. Bulk: **Delete all sample data** → modal listing counts → type `DELETE SAMPLE` → confirm → toast with Undo (10 s) before purge.
4. Dashboard sample counter reaches 0 and Launch Readiness item turns green.

Flow A4 — Create and publish a project

1. **Projects → New project**.
2. Tabs: **Basics** (title, slug auto-generated and editable with uniqueness check, short description, category, tags, date), **Case study** (rich-text sections), **Media** (cover + gallery via Media Picker, drag to reorder, alt text required), **Links & Details** (role, tools, live demo, source — URL validation), **SEO** (title, description, OG image with live preview), **Publish** (status, schedule date/time, featured toggle).
3. Autosave as Draft every 10 s; unsaved-changes guard when leaving.
4. **Preview** opens the public template with draft data (preview token, noindex).
5. **Publish** (or **Schedule**) → validation list (missing cover, missing alt text, empty description) must pass → success toast with “View on site” link.
6. Later edits create **revisions**; **History** tab allows compare and restore.
7. Drag-and-drop reorder from the list view saves immediately with Undo.

Flow A5 — Manage intro video

1. **Intro** screen shows current video, poster, captions status, enabled toggle.
2. Upload desktop MP4 (progress bar, size/type validation, compression tip), optional mobile MP4, poster, captions (.vtt).
3. Set focal point on the poster (click to choose `object-position`), end colour for transition, timeout seconds, "skip-preference default".
4. Edit scripts (English / Roman Urdu) used for the transcript.
5. **Test** buttons: *Preview intro*, *Simulate video error*, *Simulate autoplay blocked*, *Reduced-motion preview*.
6. **Save & publish**. Disabling the toggle makes the public site skip the intro.

Flow A6 — Handle an inquiry

1. Dashboard badge shows **New: 3**; open **Inbox**.
2. List with search, status filter, date sort; unread in bold.
3. Open an inquiry → shows fields, service, budget, timeline, message, received time, notification status (sent/failed).
4. Actions: **Reply** (opens `mailto:` with subject), **WhatsApp** (if number given), change status (New → In progress → Replied → Closed), **Mark spam**, add **private note**, **Delete** (Owner only, confirm).
5. Bulk select → change status/delete; **Export CSV**.

Flow A7 — Settings, navigation and flags

1. **Settings:** name, title, bios, portrait, availability toggle + text, contact email, WhatsApp, booking URL, socials, default theme, accent (from approved palette that passes contrast).
2. **Navigation & Sections:** drag to reorder homepage sections and nav items; toggle visibility; edit CTA label/target.
3. **Feature flags:** toggles for blog, testimonials, FAQ, pricing, command palette, language switcher, analytics, 3D hero, custom cursor.
4. Save → public site reflects on next load (cache revalidated); **View site** link opens in a new tab.

Flow A8 — Media library

Upload (multi-file, drag-drop) → auto-resize/WebP for images → required alt text prompt → grid/list view with search and type filter → click item for details: URL, size, dimensions, **Used in** list → Replace file (keeps URL) / Delete (blocked with explanation if in use).

Flow A9 — Analytics

Choose range (7/30/90 days) → charts for views and sessions, top projects, CTA clicks, intro play/skip/complete rates, form starts vs submits, referrers → Export CSV. Empty state when no data yet.

Flow A10 — Users and roles (Owner only)

Invite editor by email → invitee sets password → role **Editor** → Owner can change role, disable or remove. Editors do not see Users, Danger Zone, Settings > flags.

Flow A11 — Backup and restore

Export all content (JSON) → file downloads. **Import**: choose file → validation → diff preview (create/update/skip counts) → confirm → applied; a pre-import snapshot is created automatically.

Flow A12 — Danger zone (Owner only)

Delete all inquiries, reset to sample data, purge analytics. Each needs type-to-confirm and is written to the audit log.

Flow A13 — Admin error and edge cases

Case	Behaviour
Session expired mid-edit	Modal “Sign in again” keeps form state; after login the save retries.
Two admins edit the same item	`updated_at` check on save; conflict dialog with “Reload / Overwrite / Copy my changes”.
Upload fails	Row shows error + Retry; other uploads continue.
Slug already exists	Inline error with suggestion `slug-2`.

Case	Behaviour
Editor tries owner-only page	“You don’t have access” screen; RLS blocks the data anyway.
Publish with sample images	Warning, can proceed after confirmation.
Offline	Banner “You’re offline — changes will not be saved”; save buttons disabled.

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5. Page-Level Screen Inventory (public)

Screen	Key states to build
Intro overlay	loading, playing, paused, muted/unmuted, captions on/off, error fallback, autoplay-blocked, transitioning
Navbar	top, compact, hidden-on-scroll, mobile open, focus states
Hero	pre-reveal, revealed, reduced-motion
Projects grid	loading, results, empty, error, load-more
Project detail	loading, loaded, 404
Contact	pristine, invalid, submitting, success, server error, rate-limited, demo-mode
Resume	with PDF, without PDF
Footer	links, replay intro, theme toggle, socials, copyright

6. Acceptance Criteria for Flows

A flow passes when: every step above works with mouse, keyboard and touch; each alternate path has been exercised manually or by an E2E test; no step requires a page reload to recover; and nothing in the flow depends on a button that has no real function.